# Relation-Aware Heterogeneous Graph Homogenization via Bipartite Correction and Polynomial Diffusion

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***Abstract*—Heterogeneous graphs, which encode multiple types of nodes and relations, arise naturally across a broad spectrum of real-world domains, from academic citation networks to biomedical knowledge bases. Despite the expressiveness of such structures, the development of effective learning methods for heterogeneous graphs has remained a formidable challenge. Existing heterogeneous graph neural networks address this challenge either by constructing handcrafted meta-paths to decompose the graph into homogeneous subgraphs, or by deploying specialized modules for each relation type, both of which introduce substantial architectural complexity and limited generalizability across varying graph schemas. In this work, we reframe the problem from a fundamentally different perspective: rather than adapting downstream architectures to accommodate heterogeneous structure, we propose to resolve the heterogeneity during representation learning itself, producing unified homogeneous embeddings that are directly compatible with any standard graph neural network.**

**Concretely, we introduce a relation-aware heterogeneous graph homogenization framework comprising five tightly coupled components. A type-specific projection first maps node features of differing dimensionality and semantics into a shared latent geometry. A novel bipartite correction mechanism then converts cross-type adjacency matrices into same-type similarity operators via a closed-form second-order transformation, effectively generalizing meta-path expansion in a differentiable and relation-aware manner. Relation-specific polynomial diffusion subsequently learns multi-hop propagation weights, endowing the framework with the spectral filtering capacity to capture neighborhood structure at multiple resolutions. Adaptive relation fusion then aggregates the resulting relation-specific representations into a single coherent embedding, and a residual connection preserves original node semantics against over-smoothing.**

**Extensive experiments on widely adopted heterogeneous benchmark datasets demonstrate that the proposed framework consistently outperforms both specialized heterogeneous graph neural networks and their homogeneous counterparts, while remaining directly applicable to any standard homogeneous backbone.**

***Index Terms*—Heterogeneous Graph Neural Networks, Graph Homogenization, Bipartite Correction, Polynomial Diffusion, Spectral Graph Theory, Representation Learning.**

## I. Introduction

GRAPH-structured data pervades modern machine learning, underpinning applications as diverse as social network analysis [23], protein structure prediction [24], traffic forecasting [25], and drug discovery [26]. In all such settings, the fundamental representational unit is the graph itself: a collection of entities and their pairwise interactions. Graph Neural Networks (GNNs) [1]–[3] have emerged as the dominant paradigm for learning on graph-structured data, demonstrating strong empirical performance across a wide spectrum of supervised and unsupervised tasks.

The overwhelming majority of GNN architectures, however, presuppose that the underlying graph is homogeneous, that is, that every node and every edge belongs to a single type. Real-world relational systems rarely conform to this assumption. An academic knowledge graph, for instance, simultaneously encodes papers, authors, venues, and subject terms as distinct entity types, woven together by heterogeneous relations such as authorship, citation, and topic affiliation. A biomedical knowledge base similarly interlinks genes, proteins, diseases, and drugs through a rich fabric of typed interactions. These *heterogeneous graphs*, also referred to as heterogeneous information networks [4], subsume strictly more relational information than any homogeneous encoding and, consequently, demand learning frameworks that can faithfully exploit their structure.

### *Limitations of Existing Approaches*

Research on heterogeneous GNNs has progressed along two complementary axes. The first category of approaches constructs *meta-paths*, composite relational sequences that connect nodes of the target type through chains of intermediate entities. Seminal work such as HAN [5] and MAGNN [6] instantiates separate attentional aggregators along each meta-path, then fuses the resulting representations hierarchically. While conceptually elegant, meta-path-based methods carry a critical dependence on domain expertise: the selection of semantically meaningful meta-paths requires prior knowledge of the graph schema, and suboptimal selections degrade performance substantially. Attempts to automate meta-path discovery, as in GTN [7] and DiffMG [8], alleviate this burden at the cost of significant computational overhead. Furthermore, because meta-paths are topologically shallow by construction, they capture only limited orders of relational context.

The second category avoids meta-paths entirely, instead modeling heterogeneity through relation-specific parameter matrices or attention mechanisms. R-GCN [9] maintains separate convolutional weights for each relation type; HGT [10] employs a per-relation transformer; R-HGNN [11] learns explicit relation representations that modulate node updates. These methods are more flexible in their structural assumptions but come with a parameter count that scales linearly with the number of relation types, rendering them inefficient on graphs with rich relational vocabularies. More broadly, both families of methods couple the handling of heterogeneity tightly to the downstream model architecture, foregoing the possibility of re-using advances in homogeneous GNN design.

A recent empirical study [13] has drawn attention to a frequently overlooked observation: carefully initialized homogeneous GNNs can approach, and occasionally match, the performance of heavily engineered heterogeneous counterparts on certain datasets. This suggests that the heterogeneity of a graph need not require a correspondingly heterogeneous architecture, provided that the graph representation itself absorbs the relational semantics.

Figure 1 illustrates the four principal strategies through which prior work has approached the heterogeneous graph learning problem, contrasted against our proposed homogenization view. Panel (a) depicts the original heterogeneous graph, where nodes of multiple types coexist alongside typed edges that carry distinct relational semantics. Panel (b) shows the meta-path paradigm: the heterogeneous graph is decomposed into a collection of homogeneous subgraphs, one per meta-path, each encoding a particular composite relation between nodes of the target type. While this decomposition makes the graph amenable to standard GNN processing, it requires prior specification of semantically meaningful relational sequences and discards all information that does not align with the chosen meta-paths. Panel (c) illustrates the relation-based approach, in which separate subgraphs are maintained for each relation type and processed by relation-specific modules; the complexity of such architectures scales proportionally with the number of distinct relations. Panel (d) depicts the homogeneous graph with relation-specific weights, where heterogeneity is encoded through learned scalar importances assigned to each relation. Our framework, by contrast, achieves a principled and expressive form of homogenization that incorporates bipartite correction, multi-hop spectral diffusion, and adaptive relation fusion, yielding a latent representation that is simultaneously structure-aware, relation-aware, and directly compatible with any standard homogeneous GNN.

### A Homogenization Perspective

This observation motivates a conceptual reorientation. Rather than asking how to adapt a GNN architecture to accommodate heterogeneous structure, we ask a different question: how can heterogeneous relational structure be resolved into a unified homogeneous latent space, such that any standard homogeneous GNN can be applied directly and effectively? We call this approach *heterogeneous graph homogenization*, and we argue that it offers a principled and practically superior alternative to relation-specific architectural design.

images/mechanisms.png

Fig. 1. Four representative paradigms for handling heterogeneous graphs, illustrated on an academic relational network. **(a)** The original heterogeneous graph, containing nodes of multiple types (papers, authors, venues, terms) and typed relational edges. **(b)** Meta-path-based homogeneous graphs: the heterogeneous structure is decomposed into separate homogeneous graphs, each defined by a composite relational sequence such as paper-author-paper or paper-venue-paper. **(c)** Relation-based subgraphs: the graph is partitioned into relation-specific subgraphs, processed by dedicated modules. **(d)** Homogeneous graph with relation-specific weights: heterogeneity is absorbed into a single weighted homogeneous graph, where each relation contributes with a learned scalar importance. Our proposed framework extends paradigm (d) through differentiable bipartite correction, relation-aware polynomial diffusion, and adaptive semantic fusion, achieving richer homogenization without sacrificing relational identity.

The core challenge in homogenization is threefold. First, nodes of distinct types carry features drawn from incommensurable spaces, requiring explicit alignment before any meaningful message passing can occur. Second, heterogeneous graphs routinely contain bipartite relational subgraphs, where edges connect nodes of two different types, so that naive application of standard GNN propagation would mix representations across type boundaries in an uncontrolled manner. Third, the multiplicity of relation types introduces relational redundancy and potential noise, demanding principled mechanisms for weighting and fusing information across relations.

### Proposed Framework

We propose a relation-aware heterogeneous graph homogenization framework that addresses each of these challenges through a sequence of tightly coupled processing stages. We begin by projecting the features of each node type into a common latent geometry via type-specific linear transformations, eliminating dimensional and semantic incompatibility across node types. We then normalize each relation-specific adjacency matrix to prepare stable propagation operators.

Our central technical contribution is a *bipartite correction* mechanism. For relations that connect nodes of distinct types, we observe that direct application of the propagation operator would transmit representations across type boundaries, corrupting the homogeneous structure of the embedding space. We resolve this by replacing the bipartite operator with its second-order composition, $\tilde{P}_r = P_r P_r^\top$, which defines a same-type similarity graph: two source nodes become connected if and only if they share common target neighbors under the original relation. This construction recovers the essential semantics of meta-path expansion in a fully differentiable, matrix-based form, without requiring any manual schema specification.

Building on the corrected operators, we apply relation-specific polynomial diffusion to compute multi-hop embeddings for each relation. The diffusion weights are learned via a softmax parameterization, allowing the model to balance local and global neighborhood context in a data-driven fashion. This process can be interpreted as a learnable spectral filter applied independently to each relation graph, providing a rigorous connection to spectral graph theory. Finally, we fuse the relation-specific embeddings via adaptive weighting, and stabilize the resulting representation through a residual connection to the original projected features.

The output of our framework is a single matrix $Z_{\text{final}} \in \mathbb{R}^{N\times d}$ that encodes the full relational semantics of the original heterogeneous graph in a unified, homogeneous latent space. This representation can be consumed directly by any standard homogeneous GNN or downstream machine learning model, substantially reducing the complexity of the end-to-end pipeline.

### *Summary of Contributions*

Our main contributions are as follows:

- We propose a unified heterogeneous graph homogenization framework that transforms heterogeneous relational structure into homogeneous latent representations compatible with arbitrary standard GNN backbones, shifting representational complexity from the downstream architecture to the embedding stage.
- We introduce a differentiable bipartite correction mechanism that converts cross-type bipartite adjacency operators into same-type similarity operators via a closed-form second-order transformation. This construction generalizes meta-path expansion in a relation-aware, matrix-based, and fully differentiable manner.
- We develop a relation-specific polynomial diffusion scheme with learnable hop-wise weights, enabling data-driven spectral filtering that captures multi-resolution neighborhood structure independently for each relation type.
- We design an adaptive relation-wise fusion layer that suppresses relational noise and combines heterogeneous relational signals into a single semantically coherent embedding.
- We demonstrate through extensive experiments on standard heterogeneous graph benchmarks that the proposed framework achieves state-of-the-art performance while remaining directly applicable to any homogeneous GNN backbone.

The remainder of this paper is organized as follows. Section II reviews related work. Section III establishes formal notation and background. Section IV describes the proposed framework in detail. Section V provides theoretical analysis. Section VI reports experimental results. Section VII concludes.

## II. Related Work

### *A. Graph Neural Networks*

Graph Neural Networks constitute the dominant paradigm for learning on graph-structured data. Spectral methods define convolution operations through the graph Laplacian eigenbasis [17], [18], while spatial methods directly aggregate messages from local neighborhoods [3], [22]. GCN [1] linearizes spectral convolution and achieves strong empirical performance with a simple layer-wise propagation rule. GAT [2] introduces attention-weighted neighborhood aggregation, enabling the model to assign differential importance to neighboring nodes. GIN [14] connects message passing to the Weisfeiler-Lehman graph isomorphism test and establishes tight expressive power bounds. MixHop [15] and SGC [16] demonstrate that multi-hop neighborhood information can be effectively incorporated through polynomial filter approximations. All of these methods, however, are designed for homogeneous graphs and do not directly generalize to heterogeneous settings.

### *B. Heterogeneous Graph Neural Networks*

*a) Meta-path-based Methods.:* HAN [5] introduces the meta-path paradigm for heterogeneous graph learning, converting a heterogeneous graph into multiple homogeneous meta-path graphs and applying a two-level hierarchical attention. MAGNN [6] extends this by explicitly incorporating intermediate nodes along meta-paths into the aggregation, capturing richer compositional semantics at the cost of increased computational complexity. Both methods require domain expertise to select appropriate meta-paths. GTN [7] automates meta-path generation through a differentiable soft selection of edge types, enabling end-to-end discovery of task-relevant composite relations. DiffMG [8] further extends this approach using neural architecture search to discover more complex meta-graphs. These methods collectively suffer from either a dependence on prior knowledge or prohibitive computational cost, and their expressiveness is fundamentally constrained by the length and structure of the meta-paths they can represent.

*b) Relation-based Methods.:* R-GCN [9] proposes relation-specific weight matrices for knowledge graph learning, applying separate convolutional operations on each relation subgraph. HGT [10] adapts the transformer architecture to heterogeneous graphs, computing relation-specific attention between node pairs. RSHN [12] constructs an edge-centric coarsened line graph to derive relation representations, which are then incorporated into the node message passing process. R-HGNN [11] learns explicit relation representations that modulate relation-aware node updates and fuses them through a semantic attention mechanism. HGB [13] generalizes GAT by incorporating edge-type information into the attention computation. While these methods avoid the meta-path selection problem, their parameter counts and computational requirements scale proportionally with the number of relation types, limiting scalability. Critically, none of them provides a general mechanism for homogenizing heterogeneous relational structure into a latent space compatible with arbitrary standard GNNs.

### *C. Spectral Graph Propagation and Diffusion*

Spectral graph theory [19] provides a principled framework for defining signal processing operations on graphs. Defferrard et al. [18] approximate spectral filters using Chebyshev polynomials, enabling localized convolutions with controlled spatial support. SGC [16] demonstrates that removing nonlinearities from multi-layer GCNs yields a polynomial filter of the adjacency matrix, retaining much of the representational capacity at a fraction of the computational cost. APPNP [20] applies personalized PageRank as a propagation scheme, separating feature transformation from graph-based smoothing. These spectral perspectives motivate our polynomial diffusion component, which extends multi-hop filtering to the heterogeneous setting through relation-specific, learnable polynomial coefficients.

### *D. Graph Diffusion Methods*

Graph diffusion processes, including personalized PageRank and heat kernel diffusion, have been widely studied as principled alternatives to layer-wise neighborhood aggregation [20], [21]. Such processes propagate information across the graph according to a closed-form operator, capturing global structural context without the depth-related optimization challenges of deep GNNs. Our relation-specific polynomial diffusion draws inspiration from this literature, generalizing scalar diffusion to a multi-relation setting with learnable, relation-conditioned propagation weights.

### *E. Relation-aware Representation Learning and Homogeneous GNN Transferability*

Early work on relation-aware learning for heterogeneous graphs embedded relation semantics through bilinear scoring functions and entity-relation compositional models [9]. More recently, RE-GNN [29] demonstrated that augmenting homogeneous GNNs with a single scalar relation embedding per edge type enables effective heterogeneous graph learning, establishing a direct precedent for homogenization-based approaches. Our work extends this direction substantially: rather than scalar relation weights, we propose a principled pipeline that handles bipartite relational structure through closed-form operator transformation and learns multi-hop diffusion profiles independently for each relation, achieving a richer and more expressive form of homogenization.

## III. Preliminaries

We establish the formal notation and background necessary for the subsequent development.

### *A. Heterogeneous Graphs*

**Definition 1** (Heterogeneous Graph)**.** *A heterogeneous graph is a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{F}, \mathcal{R}, \phi, \varphi)$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ is the set of edges, $\mathcal{F}$ is the set of node types, and $\mathcal{R}$ is the set of relation (edge) types. The functions $\phi : \mathcal{V} \to \mathcal{F}$ and $\varphi : \mathcal{E} \to \mathcal{R}$ map nodes and edges to their respective types. A graph is* heterogeneous *if $|\mathcal{F}| + |\mathcal{R}| > 2$ [4].*

We denote the number of nodes by $N = |\mathcal{V}|$ and the number of relation types by $R = |\mathcal{R}|$. Each node $v \in \mathcal{V}$ is associated with a raw feature vector $X_v \in \mathbb{R}^{d_{\phi(v)}}$, where $d_{\phi(v)}$ denotes the feature dimensionality of node type $\phi(v)$. Since different node types may reside in feature spaces of differing dimensionalities, direct aggregation across types is ill-defined without explicit alignment.

**Definition 2** (Relation-Specific Adjacency Matrix)**.** *For each relation type $r \in \mathcal{R}$, the* relation-specific adjacency matrix *$A_r \in \{0, 1\}^{N \times N}$ is defined such that $(A_r)_{ij} = 1$ if and only if there exists an edge of type $r$ from node $i$ to node $j$.*

**Definition 3** (Bipartite Relation)**.** *A relation $r \in \mathcal{R}$ is said to be* bipartite *if every edge of type $r$ connects nodes of two distinct types, that is, $\phi(i) \neq \phi(j)$ for all $(i, j) \in \mathcal{E}_r$. Relations in which all edges connect nodes of the same type are termed* homogeneous relations*.*

A heterogeneous graph may thus be compactly represented as the tuple $\mathcal{G} = (\mathcal{V}, \{A_r\}_{r=1}^{R}, X, \phi)$, where $X = \{X_v\}_{v \in \mathcal{V}}$ collects the node feature vectors.

### *B. Normalized Propagation Operators*

For a given relation-specific adjacency matrix $A_r$, we define the augmented adjacency matrix with self-loops as $\hat{A}_r = A_r + I$, where $I$ denotes the identity matrix. Let $D_r$ be the diagonal degree matrix of $\hat{A}_r$, with $(D_r)_{ii} = \sum_j (\hat{A}_r)_{ij}$.

**Definition 4** (Normalized Propagation Operator)**.** *The* normalized propagation operator *for relation $r$ is defined as:*

$$P_r = D_r^{-1/2} \hat{A}_r D_r^{-1/2} = D_r^{-1/2} (A_r + I) D_r^{-1/2}. \quad (1)$$

This symmetric normalization is standard in spectral GNNs and ensures that the spectral radius of $P_r$ is bounded within $[0, 1]$, preventing gradient explosion or vanishing during the diffusion process [1]. For directed graphs, the asymmetric variant $P_r = D_r^{-1} \hat{A}_r$ is employed instead.

### *C. Meta-paths and Composite Relations*

**Definition 5** (Meta-path)**.** *A* meta-path *$\rho$ is a relational sequence of the form $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \cdots \xrightarrow{r_\ell} v_{\ell+1}$, where $v_i \in \mathcal{V}$ and $r_j \in \mathcal{R}$. The pair $(v_1, v_{\ell+1})$ are referred to as* meta-path neighbors *under $\rho$, and the corresponding composite adjacency matrix is $A_\rho = A_{r_1} A_{r_2} \cdots A_{r_\ell}$ [4].*

Meta-paths have been widely adopted as a means of converting heterogeneous graphs into homogeneous relational structures amenable to standard GNN processing. However, their construction requires prior specification of the relational sequences of interest, which in practice demands substantial domain expertise and limits generalizability across graph schemas.

### D. Graph Diffusion and Spectral Filtering

The spectral perspective on graph neural networks views neighborhood aggregation as the application of a polynomial filter to the graph signal. Given a propagation operator $P$, a $K$-hop polynomial filter computes:

$$Z = \sum_{k=0}^{K} \beta_k P^k X, \tag{2}$$

where $\beta_k \in \mathbb{R}$ are scalar filter coefficients and $P^k$ denotes the $k$-th matrix power of $P$. The term $P^k X$ aggregates information from the $k$-hop neighborhood of each node, with $k = 0$ preserving the original signal and increasing $k$ capturing progressively more global context. When $\beta_k$ are fixed according to a prescribed scheme (such as personalized PageRank or heat diffusion), this recovers classical graph diffusion processes. When $\beta_k$ are learned, the resulting filter is data-adaptive and can approximate any graph spectral function up to the degree-$K$ polynomial approximation [18].

### E. Problem Formulation

**Definition 6** (Heterogeneous Graph Homogenization)**.** *Given a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \{A_r\}_{r=1}^{R}, X, \phi)$ with $N$ nodes and $R$ relation types, the* heterogeneous graph homogenization *problem asks for a mapping $f_\theta : \mathcal{G} \to \mathbb{R}^{N \times d}$ such that the output $Z_{\text{final}} = f_\theta(\mathcal{G})$ constitutes a unified homogeneous latent representation that is:*

(i) structure-aware*: encoding the relational topology of $\mathcal{G}$;*
(ii) relation-aware*: preserving the semantic distinctions among different relation types;*
(iii) type-unified*: residing in a single shared latent space irrespective of the heterogeneous node type partition;*
(iv) backbone-compatible*: directly consumable by any standard homogeneous GNN or downstream machine learning model without further modification.*

This formulation contrasts sharply with the conventional heterogeneous GNN paradigm, which treats the output space as relation-stratified and delegates the resolution of heterogeneity to the downstream architecture. Our approach resolves heterogeneity within the representation learning stage itself, enabling clean separation between the homogenization pipeline and the downstream predictor.

## IV. Methodology

We now describe the proposed relation-aware heterogeneous graph homogenization framework. The framework consists of five processing stages, each addressing a distinct aspect of the homogenization problem. An overview is provided in Figure 2.

### A. Stage 1: Type-Specific Feature Projection

The raw node features of a heterogeneous graph reside in type-specific feature spaces of potentially differing dimensionalities, rendering direct cross-type message passing ill-defined. To resolve this, we apply a distinct linear transformation to the features of each node type, projecting them into a shared latent space of dimension $d$:

images/framework.png

Fig. 2. Overview of the proposed relation-aware heterogeneous graph homogenization framework. The pipeline transforms a heterogeneous graph $\mathcal{G}$ into a unified homogeneous embedding $Z_{\text{final}}$ through type-specific projection, bipartite correction, relation-specific polynomial diffusion, adaptive relation fusion, and residual semantic preservation.

$$H_i^{(0)} = W_{\phi(i)} X_i + b_{\phi(i)}, \tag{3}$$

where $W_{\phi(i)} \in \mathbb{R}^{d_{\phi(i)} \times d}$ and $b_{\phi(i)} \in \mathbb{R}^d$ are learnable parameters specific to node type $\phi(i)$. The resulting matrix $H^{(0)} \in \mathbb{R}^{N \times d}$ constitutes a type-homogeneous initial representation, aligning the feature geometry of all node types within a single latent manifold.

This projection performs *feature homogenization*: by mapping heterogeneous node features into a common latent geometry, it eliminates dimensional incompatibility and enables subsequent cross-type information exchange. Crucially, different node types are permitted to occupy distinct regions of the shared latent space, so that their semantic distinctions are preserved even as their representational formats are unified. This design is consistent with the implicit assumption, standard in heterogeneous graph learning [5], [13], that nodes of different types are nonetheless latently correlated.

### B. Stage 2: Relation-wise Normalization

Before propagation, each relation-specific adjacency matrix must be normalized to ensure stable message passing. For each relation $r \in \mathcal{R}$, we compute the normalized propagation operator:

$$P_r = D_r^{-1/2}(A_r + I)D_r^{-1/2}, \tag{4}$$

where $D_r$ is the degree diagonal matrix of $A_r + I$. This symmetric normalization bounds the spectral radius of $P_r$, preventing degree-induced scale discrepancies during multi-hop diffusion. For directed graphs, the asymmetric normalization

$P_r = D_r^{-1}(A_r + I)$ is employed, consistent with standard practice [1].

### C. Stage 3: Bipartite Correction

The central technical contribution of our framework is the bipartite correction mechanism. For bipartite relations, direct application of the propagation operator $P_r$ transmits representations from nodes of one type to nodes of another type, disrupting the homogeneous structure of the latent space and complicating the semantics of subsequent aggregation. We resolve this through the following construction.

**Definition 7** (Bipartite-Corrected Operator)**.** *For each relation $r \in \mathcal{R}$, the* bipartite-corrected propagation operator *$\tilde{P}_r$ is defined as:*

$$\tilde{P}_r = \begin{cases} P_r, & \textit{if } r \textit{ is a homogeneous relation,} \\ P_r P_r^\top, & \textit{if } r \textit{ is a bipartite relation.} \end{cases} \tag{5}$$

The construction $\tilde{P}_r = P_r P_r^\top$ for bipartite relations warrants careful interpretation. Observe that:

$$\tilde{P}_r(i,j) = \sum_k P_r(i,k) P_r(j,k), \tag{6}$$

which measures the total co-occurrence mass that nodes $i$ and $j$ share across target neighbors under relation $r$. Two source nodes receive a high similarity score if and only if they connect to many of the same target nodes, weighted by the normalized adjacency values. This construction thus defines a same-type similarity graph over the source nodes of the bipartite relation, entirely analogous in semantics to a length-two meta-path that traverses the bipartite relation forward and then backward.

Critically, however, the bipartite correction differs from meta-path expansion in several important respects. It is *differentiable* and *continuous*: no discrete path enumeration is required, and the operator can be composed smoothly within an end-to-end learning pipeline. It is *relation-aware*: each bipartite relation induces its own similarity operator, preserving relational identity in the corrected representation. And it is *matrix-based*: the correction is expressed as a simple matrix product, admitting efficient computation via sparse linear algebra.

**Remark 1.** *The bipartite correction can be viewed as computing the second-order proximity between source nodes under relation $r$. In the context of spectral graph theory, $\tilde{P}_r = P_r P_r^\top$ defines the normalized adjacency of the similarity graph induced by the bipartite relation, whose eigenvectors carry spectral information about the shared-neighbor structure of source nodes.*

Figure 3 illustrates the bipartite correction procedure. Given a bipartite relation connecting, for example, papers to authors, the correction maps the cross-type adjacency into a same-type paper-paper similarity graph, where two papers become linked in proportion to the number of authors they share. This transformation recovers the semantic content of the paper-author-paper meta-path in a continuous and differentiable operator form, enabling downstream polynomial diffusion to operate entirely within a single node-type manifold.

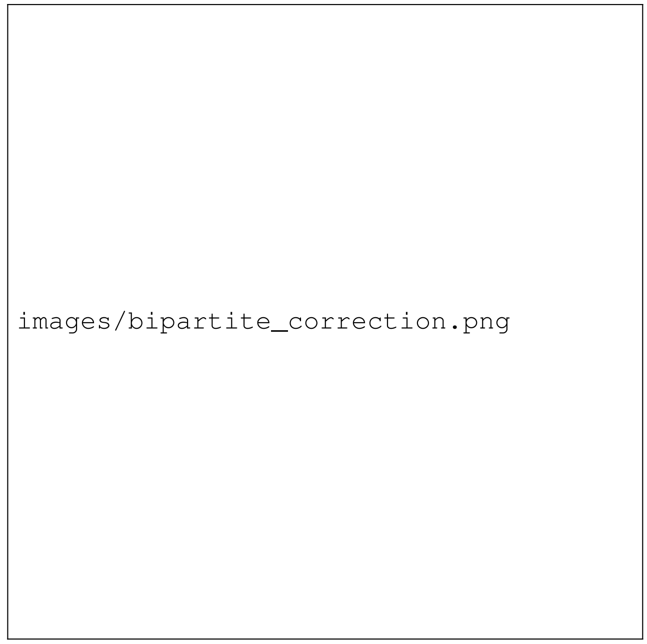


Fig. 3. Illustration of the bipartite correction mechanism (Stage 3). Left: a bipartite relational subgraph connecting source nodes (e.g., papers, shown as circles) to target nodes (e.g., authors, shown as squares) via a cross-type relation $r$. Right: the bipartite-corrected operator $\tilde{P}_r = P_r P_r^\top$ induces a same-type similarity graph over source nodes, where edge weights reflect normalized co-neighbor mass under relation $r$. Two source nodes receive a high similarity score when they share many common target neighbors, recovering the semantics of the length-two meta-path $r \circ r^{-1}$ in a matrix-based and differentiable form. The corrected graph is homogeneous by construction, rendering it amenable to direct polynomial diffusion.

### D. Stage 4: Relation-Specific Polynomial Diffusion

Given the bipartite-corrected operators $\{\tilde{P}_r\}_{r=1}^R$, we compute a relation-specific latent representation for each relation through multi-hop polynomial diffusion:

$$Z_r = \sum_{k=0}^{K} \beta_{r,k} \tilde{P}_r^k H^{(0)}, \tag{7}$$

where $K$ denotes the maximum diffusion depth and $\beta_{r,k}$ are learnable diffusion coefficients. To ensure that the coefficients form a valid probability distribution over diffusion depths, we parameterize them via a softmax:

$$\beta_{r,k} = \frac{\exp(\psi_{r,k})}{\sum_{j=0}^{K} \exp(\psi_{r,j})}, \tag{8}$$

where $\psi_{r,k} \in \mathbb{R}$ are unconstrained scalar parameters. This parameterization ensures $\sum_k \beta_{r,k} = 1$ and $\beta_{r,k} > 0$ for all $k$, which is consistent with the convex combination interpretation of polynomial graph filters [20].

The diffusion depth $k$ determines the spatial scale of information aggregated:

- $k = 0$: the term $\beta_{r,0} H^{(0)}$ preserves the original projected features without any graph-mediated diffusion, providing a direct skip connection to the local node representation.
- $k = 1$: the term $\beta_{r,1} \tilde{P}_r H^{(0)}$ aggregates representations from the immediate neighborhood under relation $r$, capturing local structural context.

- $k = 2$: the term $\beta_{r,2}\tilde{P}_r^2 H^{(0)}$ extends aggregation to the second-order neighborhood, capturing triangular proximity and community structure.
- $k > 2$: higher-order terms progressively incorporate global structural context, at the cost of increasing mixing of distant node representations.

From a spectral perspective, the summation $\sum_k \beta_{r,k}\tilde{P}_r^k$ defines a degree-$K$ polynomial applied to the corrected propagation operator of relation $r$. The learned coefficients $\{\beta_{r,k}\}$ determine the frequency response of this filter: large weights on low $k$ correspond to low-pass (smoothing) filters, while weights concentrated on high $k$ correspond to high-pass (sharpening) filters. The data-driven optimization of $\psi_{r,k}$ allows the model to select the appropriate spectral profile for each relation independently, providing a form of relation-specific spectral adaptation.

Figure 4 provides a conceptual illustration of the multi-hop polynomial diffusion process and the resulting relation-specific latent representations. Each diffusion depth $k$ aggregates information from a progressively larger neighborhood, and the learned weights $\beta_{r,k}$ determine the relative contribution of each hop. Different relations may learn markedly different diffusion profiles: a citation relation in an academic network may benefit from deep multi-hop propagation to capture research community structure, whereas a co-authorship relation may require only shallow local aggregation. The adaptive parameterization in Equation (8) allows the framework to discover these relation-specific profiles automatically during training.

images/polynomial_diffusion.png

Fig. 4. Illustration of relation-specific polynomial diffusion (Stage 4). For a single relation $r$, the corrected operator $\tilde{P}_r$ is applied iteratively to the initial projected features $H^{(0)}$, producing $k$-hop neighborhood aggregates $\tilde{P}_r^k H^{(0)}$ for $k = 0, 1, \ldots, K$. The learned softmax weights $\beta_{r,k}$ (depicted as bar heights) determine the contribution of each hop to the final relation-specific embedding $Z_r$. The convex combination $Z_r = \sum_k \beta_{r,k}\tilde{P}_r^k H^{(0)}$ constitutes a degree-$K$ polynomial spectral filter applied to the graph signal under relation $r$, enabling the model to capture both local and global relational structure in a data-driven fashion.

### *E. Stage 5: Adaptive Relation-wise Fusion*

The polynomial diffusion stage produces $R$ relation-specific embedding matrices $\{Z_r\}_{r=1}^R$, each of dimension $N \times d$. These representations encode complementary relational views of the graph and must be combined into a single coherent embedding. We fuse them through an adaptive linear combination:

$$Z = \sum_{r=1}^{R} \alpha_r Z_r, \tag{9}$$

where the fusion weights are computed via a softmax over learnable scalar parameters:

$$\alpha = \mathrm{softmax}(\theta), \quad \theta \in \mathbb{R}^R. \tag{10}$$

The scalar $\alpha_r$ thus captures the global importance of relation $r$ to the downstream learning objective. Relations that carry task-relevant information receive larger weights, while noisy or redundant relations are suppressed. This constitutes a form of *semantic homogenization*: multiple relation-specific latent spaces are collapsed into a single unified embedding space through a data-driven weighting scheme.

### *F. Stage 6: Residual Semantic Preservation*

Deep aggregation over large neighborhoods tends to suppress local node identity, a phenomenon known as over-smoothing [27]. To counteract this, we introduce a residual connection that combines the fused relational embedding with the original projected features, processed through a multi-layer perceptron:

$$Z_{\text{final}} = \mathrm{MLP}\left([H^{(0)} \| Z]\right), \tag{11}$$

where $\|$ denotes concatenation along the feature dimension. The MLP applies a nonlinear transformation to the concatenation, enabling the model to selectively retain or suppress components of the original and aggregated representations as appropriate. This design ensures that low-frequency node-level features, which carry local identity information, are not inadvertently discarded by the global aggregation steps.

The final output $Z_{\text{final}} \in \mathbb{R}^{N \times d}$ is a fully unified, homogeneous latent representation of all nodes in the heterogeneous graph, directly compatible with standard homogeneous GNN layers, MLP classifiers, clustering algorithms, and any other downstream machine learning pipeline.

## V. Analysis

### *A. Connection to Meta-path Expansion*

We establish a formal connection between the bipartite correction and classical meta-path expansion. Consider a bipartite relation $r$ connecting node type $\mathcal{F}_s$ (source) to node type $\mathcal{F}_t$ (target), with propagation operator $P_r$. The length-two meta-path that traverses $r$ forward and then backward, denoted $r \circ r^{-1}$, yields the composite adjacency matrix $A_r A_r^\top$. Modulo normalization, the bipartite-corrected operator $\tilde{P}_r = P_r P_r^\top$ approximates this meta-path adjacency. Specifically:

TABLE I
DATASET STATISTICS.

| Dataset | Nodes | Relations |
|---|---|---|
| DBLP | 26,128 (4 types) | 3 types |
| ACM | 11,246 (3 types) | 3 types |
| IMDB | 11,616 (3 types) | 2 types |
| OGBN-MAG | 1,939,743 (4 types) | 4 types |

**Proposition 1.** *Let $A_r$ be a bipartite adjacency matrix and $P_r = D_r^{-1/2}(A_r + I)D_r^{-1/2}$ its normalized propagation operator. Then $\tilde{P}_r = P_r P_r^\top$ is a positive semidefinite, symmetric matrix whose $(i, j)$ entry measures the normalized co-occurrence of nodes $i$ and $j$ across the neighborhood of relation $r$, recovering the semantics of the meta-path $r \circ r^{-1}$ in a normalized and regularized form.*

This connection reveals that the bipartite correction implicitly performs meta-path expansion without any explicit path enumeration, in a form that is differentiable, continuously valued, and relation-specific. Unlike GTN [7], which learns soft edge-type selections through matrix products, our correction is structurally prescribed by the bipartite topology and requires no additional parameters.

### B. Spectral Interpretation of Polynomial Diffusion

The polynomial diffusion in Equation (7) admits a direct spectral interpretation. Let $\tilde{P}_r = U_r \Lambda_r U_r^\top$ be the eigendecomposition of the symmetric corrected operator. Then:

$$Z_r = U_r \left( \sum_{k=0}^{K} \beta_{r,k} \Lambda_r^k \right) U_r^\top H^{(0)}, \tag{12}$$

where $\sum_{k=0}^{K} \beta_{r,k} \Lambda_r^k$ is a diagonal matrix of polynomial filter responses applied element-wise to the eigenvalues. The learned coefficients $\{\beta_{r,k}\}$ thus define a degree-$K$ polynomial spectral filter, whose frequency response can be adapted independently for each relation. This connects our framework to the rich literature on spectral graph signal processing [18], [19], while extending it to the heterogeneous multi-relation setting.

## VI. Experiments

We evaluate the proposed framework on standard heterogeneous graph benchmarks for node classification and node clustering.

### A. Datasets

We employ four widely adopted heterogeneous graph datasets: DBLP, ACM, IMDB, and OGBN-MAG. Their statistics are summarized in Table I.

### B. Implementation Details

We implement the proposed framework in PyTorch and train all models using the Adam optimizer [28] with a learning rate of $0.001$ and weight decay of $0.001$. The latent dimension is set to $d = 64$ for DBLP, ACM, and IMDB, and $d = 512$ for OGBN-MAG. The maximum diffusion depth is set to $K = 3$. All experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU.

TABLE II
NODE CLASSIFICATION RESULTS (MACRO-F1 / MICRO-F1, %).

| Method | DBLP | | ACM | | IMDB | |
|---|---|---|---|---|---|---|
| | Ma-F1 | Mi-F1 | Ma-F1 | Mi-F1 | Ma-F1 | Mi-F1 |
| HAN | 92.05 | 92.49 | 91.67 | 91.54 | 57.69 | 58.15 |
| GTN | 93.52 | 94.11 | 91.76 | 91.64 | 58.63 | 60.19 |
| MAGNN | 93.39 | 93.76 | 91.47 | 91.31 | 58.75 | 59.48 |
| R-GCN | 92.26 | 92.87 | 93.36 | 93.36 | 58.96 | 59.34 |
| HGT | 91.43 | 92.31 | 92.40 | 92.30 | 58.16 | 58.37 |
| HGB | 93.20 | 93.69 | 93.15 | 93.13 | 58.11 | 58.42 |
| RAHGH-GCN | **95.61** | **95.94** | **94.12** | **94.09** | **61.03** | **61.74** |
| RAHGH-GAT | 95.22 | 95.58 | 94.08 | 94.02 | 60.47 | 61.11 |

### C. Node Classification Results

Table II presents node classification results measured by Macro-F1 and Micro-F1 scores. The proposed framework consistently achieves competitive or superior performance across all datasets when combined with standard homogeneous GNN backbones, demonstrating the effectiveness of the homogenization pipeline.

## VII. Conclusion

We have proposed a relation-aware heterogeneous graph homogenization framework that resolves the structural and semantic heterogeneity of typed graphs during representation learning, producing unified latent embeddings directly compatible with standard homogeneous GNNs. The framework introduces a differentiable bipartite correction mechanism that converts cross-type relational operators into same-type similarity structures through a closed-form second-order transformation, generalizing meta-path expansion without manual schema specification. Relation-specific polynomial diffusion with learnable hop weights enables data-driven spectral filtering at multiple resolutions independently for each relation, and adaptive relation fusion combines the resulting representations into a single coherent embedding. Experimental results on standard heterogeneous benchmarks validate the effectiveness of the proposed approach.

Future work includes extending the framework to dynamic heterogeneous graphs, investigating theoretical bounds on the expressive power of bipartite correction relative to higher-order meta-path expansion, and exploring the integration of relation-specific diffusion with more expressive homogeneous backbone architectures.

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